A-Level Physics: Synoptic Question on Solar Sail Propulsion

Synoptic Question: The Engineering and Physics of Solar Sails

Solar sails are a form of spacecraft propulsion that rely on the radiation pressure exerted by sunlight. This question explores the integration of photon momentum, circular motion, materials science, and thermal physics in the design of a solar sail for a mission to the outer solar system.

Data and Constants

- Speed of light, $c = 3.00 \times 10^8$ m s^{-1}
- Solar Luminosity, $L_\odot = 3.83 \times 10^{26}$ W
- Gravitational constant, $G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}
- Mass of the Sun, $M_\odot = 1.99 \times 10^{30}$ kg
- Earth-Sun distance (1 AU), $r = 1.50 \times 10^{11}$ m
- Stefan-Boltzmann constant, $\sigma = 5.67 \times 10^{-8}$ W m^{-2} K^{-4}

Part A: Radiation Pressure and Force

A solar sail is made of a highly reflective aluminized Mylar film. When photons strike the sail, they undergo a change in momentum.

1. Show that for a perfectly reflecting surface of area A, the force F exerted by a solar flux (intensity) I is given by:
$$F = \frac{2IA}{c}$$
2. Calculate the solar intensity I at 1 AU from the Sun.
3. A square solar sail has a side length of 100 m. Calculate the maximum force exerted on the sail by sunlight at 1 AU, assuming 100% reflectivity.

Part B: Orbital Mechanics and Net Force

For a solar sail to "levitate" or move outwards from the Sun, the radiation pressure must overcome the Sun's gravitational pull.

4. The total mass of the spacecraft (sail + payload) is 150 kg. Calculate the gravitational force exerted by the Sun on the spacecraft at 1 AU.
5. Explain why the ratio of the solar radiation force to the gravitational force remains constant, regardless of the distance r from the Sun.
6. Determine the minimum area A required for the radiation force to exactly equal the gravitational force for this 150 kg spacecraft.

Part C: Materials and Thermal Physics

The sail material must be extremely thin to minimize mass, but it must also survive the intense thermal environment near the Sun.

7. The Mylar film has a thickness of 2.0 μm and a density of 1390 kg m^{-3}. Calculate the mass of a 100 m $\times$ 100 m sail.

8. The sail absorbs 10% of the incident sunlight (reflectivity = 0.90). At a certain distance from the Sun, the intensity is 2500 W m^{-2}. Assuming the sail acts as a black body and radiates heat from both its front and back surfaces, calculate the equilibrium temperature of the sail.

Task: The "Critical Loading" of a Solar Sail

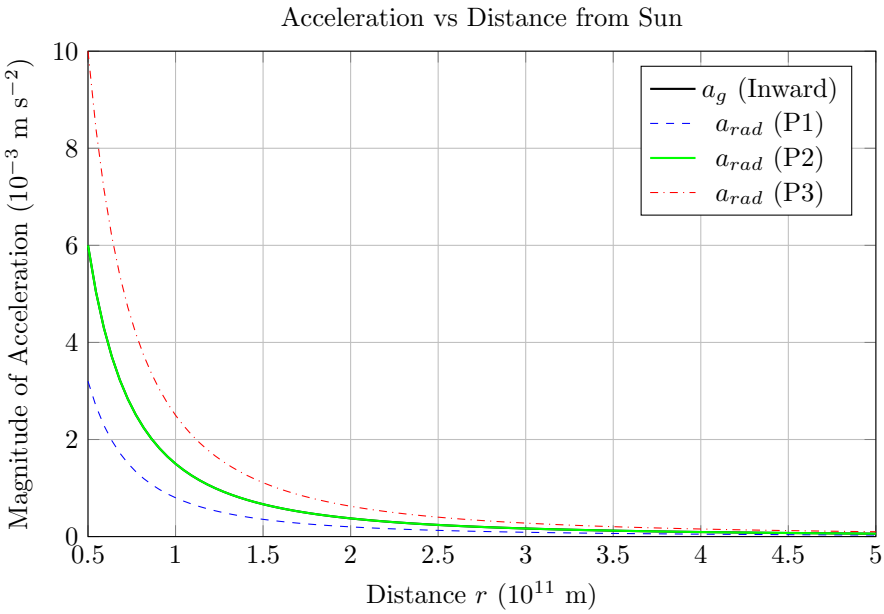
In this analysis, we define the **Net Acceleration** (a_{net}) of a spacecraft as the vector sum of the acceleration due to radiation pressure (a_{rad}) and the acceleration due to gravity (a_g).

$$a_{net} = a_{rad} - a_g = \frac{2IA}{mc} - \frac{GM_\odot}{r^2}$$

Part D: Graphical Analysis

An engineer is testing three different designs for a solar sail mission. Each design has a different "area-to-mass" ratio ($\frac{A}{m}$), measured in $\text{m}^2\ \text{kg}^{-1}$.

The graph below shows the predicted **Inward Gravitational Acceleration** (a_g) and the **Outward Radiation Acceleration** (a_{rad}) for three prototypes (P1, P2, and P3) as a function of distance r from the Sun.



1. Using the graph and the formula for a_{rad}, calculate the $\frac{A}{m}$ ratio for **Prototype 3 (P3)**.
2. Prototype 2 is said to be at "Critical Loading." Describe the motion of this spacecraft if it is released from rest at $r = 1.5 \times 10^{11}$ m.
3. A student observes that all four curves on the graph are parallel in a log-log plot. Explain why this is expected from the underlying physics.
4. In a real mission, the reflectivity (η) of the sail decreases over time due to space weathering (dust impacts and UV degradation).
   - State the effect this has on the a_{rad} curve for P3.
   - Determine the "Critical Reflectivity" value for P3 below which it will no longer be able to move away from the Sun. Assume its initial $\eta = 1.0$.

Suggested Answers and Mark Scheme

Part A: Radiation Pressure and Force

1. **(2 marks)** Momentum of one photon $p = E/c$. For reflection, $\Delta p = p - (-p) = 2E/c$.
   Force $F = \frac{\Delta p}{\Delta t} = \frac{2\sum E}{c\Delta t}$. Since Power $P = \frac{\sum E}{\Delta t}$ and $I = P/A$, then $F = \frac{2IA}{c}$.

2. **(2 marks)** $I = \frac{L_\odot}{4\pi r^2} = \frac{3.83\times10^{26}}{4\pi(1.50\times10^{11})^2} \approx 1355$ W m^{-2}.

3. **(2 marks)** $A = 100 \times 100 = 10,000$ m^2.
   $F = \frac{2\times1355\times10000}{3.00\times10^8} \approx 0.0903$ N.

Part B: Orbital Mechanics and Net Force

4. **(2 marks)** $F_g = \frac{GMm}{r^2} = \frac{(6.67\times10^{-11})(1.99\times10^{30})(150)}{(1.50\times10^{11})^2} \approx 0.885$ N.

5. **(2 marks)** Both F_{rad} and F_g follow an inverse-square law ($1/r^2$).
   $F_{rad} \propto I \propto 1/r^2$ and $F_g \propto 1/r^2$. Thus, the r^2 terms cancel in the ratio F_{rad}/F_g.

6. **(3 marks)** Set $F_{rad} = F_g \Rightarrow \frac{2(L/4\pi r^2)A}{c} = \frac{GMm}{r^2}$.
   $A = \frac{GMm(4\pi)(c)}{2L} = \frac{(0.885\times(1.5\times10^{11})^2)\times3\times10^8}{2\times(1355\times(1.5\times10^{11})^2)}$ [Simplifying constants]
   $A \approx 98,000$ m^2.

Part C: Materials and Thermal Physics

7. **(2 marks)** $V = \text{Area} \times \text{thickness} = 10000 \times 2.0 \times 10^{-6} = 0.02$ m^3.
   $m = \rho V = 1390 \times 0.02 = 27.8$ kg.

8. **(4 marks)** Power absorbed = Power radiated.
   Power absorbed per unit area $= 0.10 \times I = 0.10 \times 2500 = 250$ W m^{-2}.
   Power radiated per unit area (two sides) $= 2 \times \sigma T^4$.
   $250 = 2 \times (5.67 \times 10^{-8}) \times T^4$.
   $T^4 = \frac{250}{1.134\times10^{-7}} \approx 2.20 \times 10^9$.
   $T = \sqrt[4]{2.20 \times 10^9} \approx 216$ K.

Suggested Answers and Mark Scheme

Part D: Analysis

1. **(3 marks)** At $r = 1.0 \times 10^{11}$ m, a_{rad} for P3 is 2.5×10^{-3} m s^{-2}.
   $I = \frac{L}{4\pi r^2} = \frac{3.83\times10^{26}}{4\pi(1.0\times10^{11})^2} \approx 3048$ W m^{-2}.
   $a_{rad} = \frac{2I}{c}\left(\frac{A}{m}\right) \Rightarrow \frac{A}{m} = \frac{a_{rad}\cdot c}{2I} = \frac{(2.5\times10^{-3})\cdot(3.0\times10^8)}{2\cdot3048} \approx 123$ m^2 kg^{-1}.

2. **(2 marks)** The net force is zero ($a_{rad} = a_g$). If released from rest, the spacecraft will remain stationary relative to the Sun (effectively "levitating") or continue at a constant velocity if it already had one. It will not be bound by a standard Keplerian orbit.

3. **(2 marks)** Both a_g and a_{rad} follow the form $a = \frac{k}{r^2}$.
   Taking logs: $\log(a) = \log(k) - 2\log(r)$.
   This is a linear equation ($y = mx + c$) with a gradient of -2. Since they all have the same exponent (-2), the lines on a log-log plot will be parallel.

4. **(3 marks)**
   - **Effect:** The a_{rad} curve will shift downwards towards the a_g curve.
   - **Calculation:** For P3, the ratio of $\frac{a_{rad}}{a_g} = \frac{2.5}{1.5} \approx 1.67$.
     The force depends on $(1 + \eta)$. For $\eta = 1$, the factor is 2.
     We need $(1 + \eta_{crit})$ such that the force drops to the level of gravity:
     $\frac{1+\eta_{crit}}{2} = \frac{1}{1.67} \Rightarrow 1 + \eta_{crit} = 1.20 \Rightarrow \eta_{crit} = 0.20$.
     If reflectivity drops below 0.20 (20%), the sail fails to overcome gravity.

Cambridge International AS & A Level Physics (9702)

Paper 4 Structured Questions
Variant 2 (Higher Difficulty Extension)

Topic Context: Solar sail propulsion and synoptic mechanics/thermal physics. **Total for this variant:** 30 marks

Instructions (CIE P4 style):

- Answer all questions.
- Show full working for calculations.
- Give final numerical answers to an appropriate number of significant figures.
- Where an explanation is requested, use clear physics reasoning and relevant principles.

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- Stefan–Boltzmann constant, $\sigma = 5.67 \times 10^{-8}$ W m^{-2} K^{-4}

Question 1: Dynamics and mission feasibility [12]

A spacecraft of total mass 220 kg carries a square solar sail of side length 170 m. The sail reflectivity is 0.88 and absorptivity is 0.12. Assume normal incidence of sunlight and that the reflected component is specular.

1.1 Show that, for a partially reflecting sail, radiation pressure force may be written in the form

$$F = \frac{(1+\eta)IA}{c}$$

where η is reflectivity, I is solar intensity, and A is sail area. [2]

1.2 At $r = 1.20 \times 10^{11}$ m from the Sun, determine:

(i) the solar intensity I, [2]

(ii) the outward radiation force on the sail, [2]

(iii) the inward gravitational force on the spacecraft. [2]

1.3 The sail is tilted so that only 78% of the radiation force contributes to radial outward acceleration. Determine the net radial acceleration, including sign and direction. [2]

1.4 Open-ended: evaluate whether this configuration is suitable for sustained outward spiral transfer to the outer Solar System. Your answer should include one quantitative argument and two qualitative physics considerations. [2]

Question 2: Thermal limits and engineering trade-offs [10]

The same sail (area from Question 1) has thickness 1.50 μm and material density 1420 kg m^{-3}. At a closer approach, the sail is at $r = 5.50 \times 10^{10}$ m. Absorptivity remains 0.12.

2.1 Calculate the mass of the sail film only. [2]

2.2 Determine the equilibrium temperature if the sail radiates as a grey body from both sides with emissivity 0.80. (Use absorbed power per unit area = emitted power per unit area.) [3]

2.3 A payload instrument must remain below 330 K, and thermal coupling makes payload temperature approximately $T_{\text{payload}} = 0.84 T_{\text{sail}}$. Decide whether the thermal requirement is met. [1]

2.4 Open-ended: suggest two engineering modifications that preserve thrust performance as much as possible while reducing thermal risk. Explain the physics of each. [4]

Question 3: Critical loading and degradation model [8]

For radial motion, model acceleration as

$$a_{\text{net}} = \frac{(1+\eta)L_\odot}{4\pi c r^2}\left(\frac{A}{m}\right) - \frac{GM_\odot}{r^2}.$$

A design has $A/m = 95.0$ m^2 kg^{-1} initially, with reflectivity degrading linearly over time:

$$\eta(t) = 0.92 - 0.015t,$$

where t is time in years.

3.1 Show that the condition for critical loading ($a_{\text{net}} = 0$) gives

$$(1+\eta_{\text{crit}})\left(\frac{A}{m}\right) = \frac{4\pi c G M_\odot}{L_\odot}.$$

[2]

3.2 Calculate η_{crit} for this spacecraft. [2]

3.3 Determine the time from launch when outward acceleration ceases, according to this model. [2]

3.4 Open-ended: discuss one limitation of this model and one realistic orbital effect that would alter the prediction. [2]

Sample Solutions and Mark Scheme
CIE P4 Style – Variant 2 only

Question 1 Mark Scheme [12]

1.1 [2]

- M1: Uses momentum change ideas: reflected part contributes factor 2, absorbed part factor 1.
- A1: Correct expression $F = \dfrac{(1+\eta)IA}{c}$.

1.2(i) [2]

$$I = \frac{L_\odot}{4\pi r^2} = \frac{3.83 \times 10^{26}}{4\pi(1.20 \times 10^{11})^2} = 2.12 \times 10^3 \text{ W m}^{-2}.$$

- M1 substitution into inverse-square relation.
- A1 $I = 2.12 \times 10^3$ W m^{-2} (allow 2.11–2.12 $\times 10^3$).

1.2(ii) [2]

$$A = 170^2 = 2.89 \times 10^4 \text{ m}^2, \quad F_{\text{rad}} = \frac{(1+0.88)(2.12 \times 10^3)(2.89 \times 10^4)}{3.00 \times 10^8} = 3.84 \times 10^{-1} \text{ N}.$$

- M1 correct area and force setup.
- A1 $F_{\text{rad}} \approx 0.384$ N.

1.2(iii) [2]

$$F_g = \frac{GM_\odot m}{r^2} = \frac{(6.67 \times 10^{-11})(1.99 \times 10^{30})(220)}{(1.20 \times 10^{11})^2} = 2.03 \text{ N}.$$

- M1 correct Newtonian gravity expression.
- A1 $F_g \approx 2.03$ N.

1.3 [2]

$$F_{\text{out,radial}} = 0.78 F_{\text{rad}} = 0.299 \text{ N}.$$

$$F_{\text{net,radial}} = 0.299 - 2.03 = -1.73 \text{ N}.$$

$$a_{\text{net,radial}} = \frac{-1.73}{220} = -7.85 \times 10^{-3} \text{ m s}^{-2}.$$

Direction: inward (towards Sun).

- M1 resolves radial component and subtracts forces with sign.

- A1 correct magnitude/sign and direction.

1.4 (open-ended) [2] Sample points for credit (any two well-developed points, one must be quantitative):

- Quantitative: inward net acceleration found, so this configuration cannot sustain outward spiral at this r.
- Larger A/m or lower total mass required for positive net outward acceleration.
- Sail attitude optimization trades radial thrust against transverse components needed for orbital energy changes.
- Degradation of reflectivity further reduces future thrust margin.

Question 2 Mark Scheme [10]

2.1 [2]

$$m_{\text{sail}} = \rho A t = (1420)(2.89 \times 10^4)(1.50 \times 10^{-6}) = 61.6\,\text{kg}.$$

- M1 correct volume/mass relation.
- A1 $m_{\text{sail}} \approx 61.6$ kg.

2.2 [3]

$$I = \frac{3.83 \times 10^{26}}{4\pi(5.50 \times 10^{10})^2} = 1.01 \times 10^4\,\text{W m}^{-2}.$$

Absorbed per unit area:

$$P_{\text{abs}}/A = 0.12I = 1.21 \times 10^3\,\text{W m}^{-2}.$$

Emitted per unit area from two sides:

$$P_{\text{emit}}/A = 2\varepsilon\sigma T^4 = 2(0.80)(5.67 \times 10^{-8})T^4.$$

So,

$$T = \left(\frac{1.21 \times 10^3}{2(0.80)(5.67 \times 10^{-8})}\right)^{1/4} = 339\,\text{K}.$$

- M1 finds local intensity correctly.
- M1 equates absorbed and emitted power with two-sided emission and emissivity.
- A1 $T \approx 339$ K.

2.3 [1]

$$T_{\text{payload}} = 0.84(339) = 285\,\text{K}.$$

Thermal requirement met ($285 < 330$ K). [1]

2.4 (open-ended) [4] Award up to [2] per justified modification; max [4]. Examples:

- Increase emissivity (especially IR emissive rear coating) while maintaining high solar reflectivity: raises radiative cooling with small penalty to photon pressure.
- Use slight attitude control and duty-cycled perihelion pointing: reduces absorbed flux peaks while preserving mission-average thrust.
- Add thermally isolating boom and low-conductivity support paths: decouples payload temperature from sail temperature.
- Multi-layer sail with reflective front and emissive back: improved thermal rejection with comparable momentum transfer.

Question 3 Mark Scheme [8]

3.1 [2] Set $a_{\text{net}} = 0$ and cancel common $1/r^2$ term:

$$\frac{(1+\eta)L_\odot}{4\pi c}\left(\frac{A}{m}\right) = GM_\odot \Rightarrow (1+\eta_{\text{crit}})\left(\frac{A}{m}\right) = \frac{4\pi c G M_\odot}{L_\odot}.$$

- M1 sets critical condition and rearranges.
- A1 obtains stated expression.

3.2 [2]

$$\frac{4\pi c G M_\odot}{L_\odot} = \frac{4\pi(3.00\times10^8)(6.67\times10^{-11})(1.99\times10^{30})}{3.83\times10^{26}} = 130.8.$$

$$1+\eta_{\text{crit}} = \frac{130.8}{95.0} = 1.376 \Rightarrow \eta_{\text{crit}} = 0.376.$$

- M1 computes constant and substitutes A/m.
- A1 $\eta_{\text{crit}} \approx 0.38$.

3.3 [2] Outward acceleration ceases when $\eta(t) = \eta_{\text{crit}}$:

$$0.92 - 0.015t = 0.376 \Rightarrow t = \frac{0.92-0.376}{0.015} = 36.3\,\text{years}.$$

- M1 uses degradation model with critical value.
- A1 $t \approx 36$ years.

3.4 (open-ended) [2] Any two valid evaluative points, e.g.:

- Limitation: assumes strictly radial motion and constant A/m; ignores sail deformation and control dynamics.
- Limitation: linear degradation of reflectivity may be unrealistic (likely non-linear and environment-dependent).

- Orbital effect: changing heliocentric speed and trajectory geometry alters effective incidence angle and hence usable radial thrust.
- Orbital effect: planetary perturbations and non-Keplerian steering maneuvers modify net acceleration history.